THE SOLUTION · RESEARCH REPORT

DESIGNING THE EXPOSURE STUDY

Per-participant KPIs, lawful collection and coding of short-form video, and a mock feed app that sequences content to test opinion change — with the ethics, debriefing and data-protection requirements such a study must meet.



Israeli sovereignty over all the land Partition and shared arrangements A single state for all, or a Palestinian state over all the land

Prepared for The Solution · game design research

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Scope Proposals documented in scholarly, legal, policy and press sources, presented descriptively. Inclusion is not endorsement.

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Compare conditions, license clips, debrief deeply

A curated short-video experiment on the Israeli–Palestinian conflict can be run lawfully and ethically, but it has to be built to detect **small differences between conditions**, not to read meaningful change off any one participant. The closest precedents are paradoxical-thinking clips shown to Jewish Israelis, YouTube-style recommendation experiments and a TikTok-clone study. At best they moved attitudes by roughly **d = 0.1–0.4**. They often moved emotions and watch time far more than opinions, and short-session effects typically **lose about half their size within weeks**. A single four-minute session should therefore be powered for about d = 0.15–0.20. With a baseline covariate, that means roughly **270–360 completers per arm in each population analysed** (computed here). Per-person "KPIs" are best treated as exposure, data-quality and welfare indicators. Individual attitude change should be flagged only when it clears a reliable-change threshold, which for a single 0–100 thermometer sits **near 31 points**. Lawful video is the hardest constraint. None of the official channels reliably supplies video files: YouTube's Researcher Program, Meta's Content Library, TikTok's Research API and the EU Digital Services Act's Article 40 all fall short. TikTok's API does not list Israel among its eligible countries, and every platform's terms forbid automated collection. The cleanest stimulus set is therefore **commissioned or licensed clips**, matched across Hebrew, Arabic and English. A separately coded reference corpus can be held under fair use (Israel, US) or under an EU partner's text-and-data-mining exception. The mock app should be a fork of an open video-native research feed (DICE-TikTok or WatchLens). Its conditions should hold topic share, runtime and slot position constant, and it should log every impression with its exposure context. On ethics, every applicable regime points to **full committee review** and **explicit consent to processing political opinions**. Political opinions are special-category data under the GDPR and "especially sensitive" data under Israel's Amendment 13. The consent form must disclose graphic content even if it conceals the persuasive aim. The study also needs trauma-aware screening and a debrief that does more than explain, because standard debriefs leave politically charged beliefs in place. The team's prototype already has the right architecture. It needs changes in measurement validity, condition balance, the timing of randomisation, telemetry accuracy, welfare controls and the debrief.

Per-person KPIs should screen quality while effects belong to conditions

"KPI" suggests a per-person score of success. The measurement literature points the other way. At the effect sizes this kind of study can plausibly produce, one person's pre/post change on a conflict attitude is mostly measurement error. The useful per-person indicators are whether the treatment was delivered, whether the data are trustworthy and whether the participant came through safely. Persuasion is an estimand defined by comparing conditions.

A post-2023 battery has to work around floor effects

Few constructs here have a single standard instrument. In Israeli and Palestinian samples, the usual practice (Halperin, Bruneau, Kteily, Hameiri) is short 2–8-item composites on 1–7 scales or 0–100 sliders, with alphas mostly between .70 and .85. These sit alongside single-item 0–100 measures: the ANES-style feeling thermometer and the "Ascent of Man" dehumanisation slider (Bruneau & Kteily 2017). **Target specificity matters.** Bruneau and Kteily had Israelis rate "Palestinians, Arabs, Muslims, Residents of Gaza, Members of Hamas, Arab-Israelis" separately, and Palestinians rate Israelis, settlers

and Israeli peace activists. In the 2014 Gaza war, each side rated the other about 37–40 points less "evolved" than its own group (Bruneau & Kteily 2017). A later JPSP supplement rewords the item as "How human are [target]?", anchored from "Not at all human" to "Completely human" (APA supplement). In the affective-polarisation literature, social-distance items tap a distinct dimension. Respondents rating an outgroup also tend to picture its elites rather than ordinary members (Druckman & Levendusky).

The post-October-2023 baseline creates severe floor effects. In the July 2024 Palestinian-Israeli Pulse, Palestinians gave Israeli Jews a mean "humanity" score of 6 out of 100, and 71% chose zero. Israeli Jews gave Palestinians 14, and 42% chose zero (Pulse, Sept 2024). Single 0–100 items will have compressed variance in such samples. Multi-item composites, and items that separate civilians from armed groups, are more likely to register change. For solution support, **copying the Pulse wording gives the only national benchmark tracked over time on both sides**. In 2024, the two-state "concept" had 21% support among Israeli Jews (42% in 2020), 40% among Palestinians and 72% among Israeli Arabs. Annexation of the West Bank without equal rights for Palestinians reached 42% among Israeli Jews, the highest the Pulse has recorded. A detailed peace package drew 25% of Israeli Jews and 34% of Palestinians. In a split-sample incentives experiment, that support rose to as much as 60% and 55% respectively (Pulse). Format changes the headline: PSR's single-choice format produced a two-state plurality that PSR itself says exceeds support measured with support/oppose items (PSR, Jan 2024).

Construct	Recommended measure	Evidence in Israeli or Palestinian samples	Role
Outgroup warmth	0–100 thermometer with ANES anchors, for several targets (e.g., Palestinians in Gaza, in the West Bank, Hamas; Israeli Jews, settlers)	Used throughout the Bruneau–Kteily work	Primary composite component
Blatant dehumanisation	Ascent of Man slider or "How human are [target]?" 0–100, scored ingroup minus outgroup	37–40-point gaps in 2014; Pulse means of 6 and 14 in 2024	Primary
Solution support	Pulse support/oppose items per framework, including the mirror options and the detailed package	National benchmarks 2020–2024	Primary and benchmark
Hope	3–4 items adapted from Cohen-Chen et al.	α = .84 among Israelis; Palestinian two-item version r = .50 (Bruneau & Kteily)	Primary or secondary
Trust	Two Nadler–Liviatan items	Reverse-keyed item failed in the Palestinian sample (Bruneau & Kteily)	Secondary
Parochial empathy	Ingroup minus outgroup compassion sliders	Ingroup and outgroup empathy had opposite, independent effects (Bruneau, Cikara & Saxe)	Secondary
Threat and fear	Pulse fear item	52% of Israeli Jews feared harm "greatly" in 2024; 78% of Palestinians worried (Pulse)	Secondary and moderator
Conflict-supporting beliefs	Ethos-of-conflict short form, or three 0–100 "unfreezing" items (α = .80)	Hameiri et al. (PNAS 2014)	Mechanism check for the paradoxical arm
Support for violence	Collective-aggression scale (8 items, α = .78)	Casualty-tolerance item hit a ceiling: mode 1,000, chosen by 49.9% (Bruneau & Kteily)	Secondary, with care
Knowledge	Short factual quiz	Games such as PeaceMaker raise knowledge reliably (Cuhadar & Kampf)	Secondary
Implicit attitudes	Omit, or keep as an immediate exploratory measure	Changes do not carry over to explicit attitudes or behaviour (Forscher et al. 2019); they fade within hours to days (Lai et al. 2016)	Exploratory only

Hebrew and Arabic psychometric validation of the thermometer and Ascent items, beyond their use in these samples, was not found. Neither were test-retest reliabilities for Israeli or Palestinian samples. The team should estimate both itself in a pilot.

Realistic effects run d = 0.1–0.3 and halve within weeks

The benchmark studies agree that durable effects of d ≈ 0.3–0.4 required repeated or intensive exposure. Single short sessions produced small effects or none, and emotions moved more readily than attitudes. **No peer-reviewed randomised experiment of TikTok or Reels exposure on Israel–Gaza attitudes was found.**

Study	Sample and dose	Attitude effect	Durability
Paradoxical thinking (Hameiri et al. 2014)	161 Jewish Israeli completers; six ~5-minute sessions of 3 paradoxical clips plus 3 ads	Compromise η^2 = .04, unfreezing η^2 = .03 (d ≈ 0.35–0.40, computed from reported means and SDs)	Compromise d ≈ 0.37 at one year (computed), with 41% retained
Group malleability (Goldenberg et al. 2018)	508 Jewish Israelis; 5-hour workshop	Negative attitudes d = 0.18 rising to 0.34; concessions d = 0.45 and trust game d = 0.64 at six months (magnitudes versus a coping-control arm)	Grew to six months
Perspective-giving (Bruneau & Saxe 2012)	Israeli–Palestinian dyads; about 15 minutes	Palestinians improved after giving their perspective; Israelis after taking the other's perspective	None left after one week
YouTube-like feed (Liu et al. 2025)	8,883 US adults; about 23 minutes	Slanted versus balanced recommendations near zero (MDE 0.02–0.04 on a 0–1 scale); the seed video alone moved about 0.04	Not analysed
TikTok clone (Incerti et al. 2026, preprint)	About 8,600 US adults; 4-minute sessions	Pro-China influencer content +0.21 on a 4-point scale; state media −0.09; about +0.1 per extra minute	Not measured
Partisan TikToks for a week (Landgrave et al.)	US students	No attitude change; sadness, anxiety and anger rose	Not measured
X feed reranking (Piccardi et al. 2025)	1,256 users; 7 days	Out-party thermometer +2.11 (95% CI 0.15–4.06) when animosity content was reduced and −2.48 when it was increased	Not measured
US megastudy (Voelkel et al. 2024)	32,059 participants; under 8 minutes each	23 of 25 treatments reduced partisan animosity	Some effects persisted at two weeks
Decay benchmark (Coppock 2017)	18 panel survey experiments	n/a	About half the effect remained after 10 days

Heterogeneity is the rule. Paradoxical thinking moderated the views of rightists and centrists most, and did little for leftists (SIPR review). Messages must be extreme "but not too extreme" or they are rejected (HUJI CRIS), and the effect depends on where a message falls relative to the recipient's latitude of rejection (JSPP 2020). A 2025 study found that a real campaign to improve Palestinians' daily lives made right-wing Jewish Israelis *less* supportive of concessions and less hopeful (BFHU summary). That bears directly on any humanising arm. In a January 2025 tournament with 1,237 Jewish Israelis, loss-

framing raised support for ending the war across the spectrum, while recategorisation worked mainly on the left (Ushomirsky et al. 2026). Political orientation and group membership must therefore be preregistered moderators.

About 270–360 completers per arm, per population, for one session

The table uses the standard normal approximation for a two-arm comparison at 80% power. The required n is multiplied by about $(1 - \rho^2)$ when a pretest correlated $\rho = .7$ with the outcome enters as a covariate. EGAP notes that prognostic pre-treatment covariates are what buy this precision (EGAP). The last column applies a Bonferroni correction for three treatment-versus-control comparisons. All figures are computed here, are per arm and are before attrition.

Target effect (d)	No covariate, α = .05	With pretest (ρ = .7)	With pretest and 3 comparisons (α = .0167)
0.40	98	50	67
0.30	174	89	119
0.20	393	200	267
0.15	698	356	475
0.10	1,570	801	1,068

Three implications follow. First, **analyse each population separately**. The outgroup for a Jewish Israeli is the ingroup for a Palestinian, so pooled "warmth toward the outgroup" mixes different estimands. A four-arm study among Jewish Israelis and Arab citizens of Israel needs roughly 1,100–1,900 completers per population. Second, follow-ups are expensive. The Israeli precedents lost 20–60% of participants within months: retention was 97%, 80% and 59% at two weeks, two months and six months in Goldenberg et al., and 41% at one year in Hameiri et al. (Goldenberg; Hameiri). If an immediate d = 0.2 halves to 0.1 by two weeks, detecting it would need roughly 800–1,070 retained per arm. That argues for treating the delayed wave as secondary unless the design uses repeated sessions. Third, a null is informative only with a stated precision. Nyhan et al. framed their result as ruling out effects of ±0.12 SD and used exploratory equivalence bounds of ±0.1 SD (Nyhan et al. 2023). Liu et al. concluded that their results reveal "practical limits to effect sizes that are feasibly detectable in academic experiments" (Liu et al.).

Individual change is noise until it clears a reliable-change bar

The Jacobson–Truax reliable change index divides a person's change by $S_{diff} = \sqrt{2} \times SD \times \sqrt{(1 - r)}$, where r is test-retest reliability. Change beyond ±1.96 is taken as exceeding measurement error (Blampied 2022). Take a thermometer with SD ≈ 25 and an assumed r = .80. **A reliable individual change must exceed about 31 points**, while plausible average treatment effects are a few points. For a single 7-point item with SD ≈ 1.5 (close to Hameiri's compromise SDs) and an assumed r = .70, the threshold is about 2.3 scale points (computed here). Nearly every individual "KPI" movement will therefore sit inside the noise band, even when the difference between condition means is clearly detectable. The practical design follows from this. Estimate r from the control arm's pre/post pair. Report per-person RCI flags descriptively. Treat the share of reliable changers in each arm as a secondary, condition-level comparison. Never rank or select individuals by their raw change.

The primary analysis should be ANCOVA or regression of the post score on condition, the baseline score and prespecified prognostic covariates, run within each population. The moderators (political orientation, identity group) should be preregistered. Pretest measurement is safe and efficient. Clifford, Sheagley and Piston found that pre-treatment outcome measures "do not bias treatment effect

estimates and greatly improve precision" (APSR 2021). A 2025 replication with N = 13,163 found only "a small attenuation" and still recommends the design (Jordan et al.). Sensitisation can be reduced further in two ways. Bruneau and Saxe hid their key items among about 50 randomised fillers (Bruneau & Saxe), and Liu et al. ran the baseline about a week before treatment (Liu et al.).

Engagement metrics are outcomes, and quality checks are pre-treatment screens

Engagement moves a lot while attitudes do not. In Liu et al., slanted recommendations shifted like-minded choices by about 6 percentage points and changed time on the platform by +7.3 or −4.9 minutes against a 23-minute average, with essentially unchanged attitudes (Liu et al.). Watch time, completion, skips and likes therefore belong in the preregistration as secondary outcomes and manipulation-delivery checks, not as proxies for persuasion. Dropping participants on post-treatment variables, controlling for post-treatment mediators or subsetting on them biases estimates (Montgomery, Nyhan & Torres). So inattention should be screened before treatment, and post-treatment checks should be reported but not used to exclude (JEPS). Watch-time dose-response analyses are themselves post-treatment and endogenous whenever participants can skip. Behavioural anchors validated in Israeli samples are mainly incentivised games, such as a dictator allocation to Palestinians or a trust game with a Palestinian confederate (Goldenberg et al.). Donation splits and petition sign-ups are cheaper, but no Israeli validation was found, and Incerti et al. saw no petition effect even where attitudes moved (Incerti et al.).

The **data-quality threat model changed in 2025**. Westwood's autonomous "synthetic respondent" evaded standard checks 99.8% of the time at about $0.05 per survey (404 Media). A critic argues the threat is overstated because the agent was purpose-built and no large-scale infiltration has been shown (Research Live). Human LLM use is better documented: an estimated 33–46% of MTurk workers on a 2023 summarisation task (Veselovsky et al.). Prolific reports 98.7% precision for its LLM checks and a 0.99% flag rate, but these are vendor figures (Prolific). Prolific also restricts the grounds for rejection. Only instructional and nonsensical-item checks count; in studies of five minutes or more a participant must fail at least two; and time limits are not valid grounds (Prolific policy). A video app with per-clip telemetry is itself a strong screen. Zero pauses, identical dwell distributions, no touch events on a phone or playback in a background tab can all be preregistered as analysis exclusions, separate from payment rejection.

KPI family	Per-person indicators	How to use them
Exposure delivered	Clips shown, foreground watch time excluding pauses, watch ratio, completions, time to skip, replays, likes	Manipulation-delivery check; preregistered secondary outcomes; never an exclusion filter
Data quality	Pre-treatment attention check, telemetry plausibility, device and touch capability, survey duration, open-text authenticity flags	Preregistered pre-treatment exclusions; everything else in sensitivity analyses
Attitude change	Change on each primary composite, with an RCI flag calibrated on the control arm	Descriptive only; inference comes from condition-level ANCOVA
Welfare	In-feed distress check, early stops, skips of flagged clips, debrief completion, post-debrief re-measure	Safety monitoring and stop rules; residual post-debrief effect reported as a safety outcome
Retention	Completion of each follow-up wave	Attrition analysis by condition

A defensible wave structure, assembled from these precedents, has four parts. Wave 0 is the baseline about a week ahead, with key items among fillers. Wave 1 is exposure, an immediate post-test and the debrief. Wave 2 comes at one to two weeks. An optional Wave 3 comes at one to three months. In-feed momentary emotion items, as in Piccardi et al., can capture the emotional effects that short video produces most reliably (Piccardi et al.).

No official channel delivers the video files a stimulus set needs

Research programmes yield metadata, cleanrooms and long queues

Each official route gives something different, and none hands over the media files a stimulus set requires.

Route	Access from Israel	What it provides	Main constraints
YouTube Researcher Program	Eligible. Israel is on the country list for accredited, not-for-profit, degree-granting institutions (YouTube)	Scaled Data API access to public video metadata	No video files. Data must be refreshed every 30 days until "fixed as to a point in time" (terms). The search endpoint favours freshness over complete retrieval (Rieder et al.)
Meta Content Library and API	Likely. Only sanctioned jurisdictions are excluded (Meta)	Reels viewable in the web tool; API with more than 100 fields	API data stays in a cleanroom with no local download. CSV export only for "widely-known" accounts. Budget of 500,000 results per seven days. Meta says there are no fees, while Bellingcat, citing SOMAR, reports $1,000 setup plus $371/month for ICPSR enclaves (Bellingcat)
TikTok Research API	Not eligible. Applicants must be at institutions in the US, EEA, UK, Canada or Switzerland (TikTok)	Metadata including a `voice_to_text` transcript; no media URL (codebook)	2023 terms required refresh-and-delete every 15 days and a courtesy copy 30 days before publication (Tech Policy Press). A Slovak team waited 21 months for working access (KInIT)
DSA Art. 40(4), vetted researchers	Requires a research organisation under EU law, which in practice means an EU partner; non-EU status unclear (Commission FAQ)	Data transmission or a secure processing environment	80 working days for the Digital Services Coordinator to act. As of July 2026 there had been 77 applications and "no known successes", according to an uncited author figure in the DSA Observatory
DSA Art. 40(12), public data	No vetting by a Digital Services Coordinator; non-profits included (Tech Policy Press)	Public data, potentially via scraping	A joint request by about 20 organisations had not been granted by any platform (DSA Observatory)

The DSA regime is legally real but practically immature. The Commission says the delegated act entered into force on 29 October 2025 (Commission FAQ); one law-firm summary gives 21 July 2025 (VBB). The Commission preliminarily found TikTok and Meta in breach of their public-data access duties on 24 October 2025 (European Commission). The regime's main advantage for reproducibility is Recital 29: providers "should not impose archiving, storage, refresh and deletion requirements" unless the Digital Services Coordinator specifies them (Springlex). The team should file these applications early through partners, but **the study design must not depend on them**.

Copyright covers analysing clips more readily than showing them

Platform terms and copyright are separate risks. **All three platforms' consumer terms ban automated collection without written permission.** YouTube also bans downloading except "as expressly authorized by the Service" (YouTube ToS). Its API policies forbid storing copies of audiovisual content without written approval (III.E.1.a) (developer policies). Instagram bars collecting information "in an automated way without our express permission" (Instagram). TikTok's licence to viewers covers entertainment use only, but it states that "nothing in these Terms affects any rights you are given by

law" (TikTok ToS). In *Meta v. Bright Data* (N.D. Cal., January 2024), the court held that Meta's terms did not reach logged-off scraping of public data. Meta then dropped the case, so the ruling is not appellate precedent (Farella Braun + Martel). In *Sandvig v. Barr* (D.D.C. 2020), the court held that breaching terms for academic research is not a crime under the US Computer Fraud and Abuse Act (ACLU). For an Israeli team, the realistic exposure from breaching terms is account bans, IP blocking or a contract claim. No Israeli case law on research scraping was found.

Jurisdiction	Research route	Analysing a corpus	Showing clips to participants	Watch-outs
Israel	Copyright Act 2007 s.19 fair use, open-ended and naming "research" (translation); a 2022 Ministry of Justice opinion treats ML training as generally fair use (Herzog Fox & Neeman)	Strong, given the transformative, non-expressive use	Factor-based: short excerpts, a closed participant pool and no redistribution help	The s.22 incidental-use exception does not cover deliberately included music; no Israeli decision applying s.19 to research datasets or stimuli was found; the MoJ opinion is not binding
United States	17 U.S.C. §107 fair use (LII); DMCA §1201 exemption for text and data mining of films by non-profit higher-education researchers (renewal petition)	Strong under fair use	Factor-based	Whether §1201 covers streamed YouTube content is unverified
European Union	DSM Directive Art. 3 text and data mining by research organisations with lawful access; copies may be retained "for the verification of research results"; Art. 7 makes contrary contract terms unenforceable (EUR-Lex)	Strong, and immune to platform terms	Not obviously covered; rests on national implementations of InfoSoc Art. 5(3)(a), "illustration for... scientific research" (EUR-Lex)	In November 2024 the Hanseatic Higher Regional Court treated YouTube's "rolling cipher" as an effective technical protection measure (TorrentFreak)
United Kingdom	CDPA s.29A computational analysis for non-commercial research; contract override unenforceable (legislation.gov.uk)	Strong	No; transferring a copy to anyone else infringes	Narrow purpose limit

Embedding avoids reproduction but gives up experimental control. YouTube forbids modifying or blocking any part of its player and separating audio from video (III.I.6–7) (developer policies). Instagram's oEmbed is "only meant to be used for embedding", requires app review and documents no playback-event API (Meta). An embedded clip also cannot be trimmed, captioned uniformly or stripped of engagement counts, and it disappears if the creator deletes it mid-study.

Commissioned or licensed clips solve law and stimulus equivalence together

The cleanest route for stimuli is to **commission, license or obtain creator permission for every clip shown to participants**. Text-and-data-mining exceptions do not obviously cover showing clips to people, and Israeli and US fair use for whole clips with licensed music is untested. Commissioning also solves a design problem that downloads cannot. It lets the team match Hebrew, Arabic and English versions on length, production quality and speaker type, and it removes engagement counters and the platform branding that signals source credibility. The DICE-TikTok README itself points researchers toward "your institution's policies" before using `yt-dlp` for stimuli (DICE-TikTok). Real platform clips still have a place. They form a **reference corpus** that defines what "typical" Reels, Shorts and TikToks

on the war look like. Licensed stimuli can then be benchmarked against it, and a subset of real clips can be shown where fair-use factors are strong: short excerpts, a closed pool, no redistribution and no edits that misrepresent creators. Each stimulus should carry its rights basis as a metadata field.

Collection tools rank by legal exposure, and all of them break

From lowest to highest legal risk, the options are: official APIs and the Meta Content Library; embedding through oEmbed; logged-off browser capture of public pages; the Zeeschuimer extension feeding 4CAT, with a research account; command-line downloaders without login; and logged-in downloaders or tools that circumvent YouTube's cipher, which carry EU anti-circumvention exposure. Zeeschuimer captures what a researcher scrolls in Firefox, covering TikTok posts and comments and Instagram posts, including the Reels page (Bellingcat). From version 2025.11.12, `yt-dlp` needs an external JavaScript runtime such as Deno for full YouTube support (yt-dlp issue), and YouTube's newer "PO token" measures may defeat even that setup (GIGAZINE). Instaloader is heavily rate-limited when used anonymously and risks account bans when logged in (Bellingcat). For preservation, Bellingcat's Auto Archiver (v1.2.0, January 2026) writes WACZ web archives with verifiable manifests and has preserved more than 150,000 items (Bellingcat). Meta's open-source PDQ and TMK+PDQF perceptual hashes match the near-duplicate re-uploads that are common in war content (Meta). The Berkeley Protocol sets chain-of-custody norms for this kind of material (OHCHR). Every tool broke repeatedly in 2025–2026, so the team should pin versions, log failures and avoid relying on a single tool.

Field group	Fields to keep for every item
Identity and provenance	Canonical URL and platform ID; account handle and ID, account type and verification status at collection; upload time (UTC); collection time; collector, tool and version
Content	Caption, hashtags, mentions, audio or music ID, duration, language, region if provided
Engagement snapshot	View, like, comment and share counts, with the time each was observed
Platform labels	State-affiliated, AI-generated and sensitive-content interstitial flags (the TikTok codebook is a useful minimum schema (TikTok))
Integrity	SHA-256 of the media file; PDQ/TMK perceptual hash; WACZ archive reference
Derived	Transcript, OCR text and keyframes, each with model name and version
Status	Availability at each re-check (live, removed, restricted)
Rights and ethics (added here)	Rights basis (licence, permission, fair use, TDM); graphicness rating; ethics-committee approval status

The corpus itself is sensitive personal data

Videos of identifiable people stating positions on the war reveal "political opinions". That makes them special-category data under GDPR Art. 9, with a research route through Art. 9(2)(j) and Art. 89 safeguards (GDPR Art. 9). They are also "especially sensitive" data under Israel's Privacy Protection Law as amended by Amendment 13, in force since 14 August 2025, though some sources give 15 August (DLA Piper). No research exemption appears in English-language summaries of the Israeli law. The Privacy Protection Authority's April 2025 draft AI guidance would treat unauthorised scraping as a potential "serious security incident" and questions collecting social-media personal data without clear lawful consent. Whether it has been finalised is unconfirmed (Gornitzky). Large-scale scraping of identifiable private users is therefore the highest-risk scenario. Prioritising news, organisational, state and public-figure accounts lowers the risk. Wartime footage often shows victims, including children, who never chose to be filmed, and ethics review should address them separately from the uploader.

Retention can be tiered. API-derived metadata follows each programme's refresh rules. Self-collected media held under a copyright exception can be kept for verification under restricted access; DSM Art. 3(2) allows this, while UK s.29A bars transfer. Personal data should be pseudonymised, with handles held separately from codes. Removal is common and not random. Human Rights Watch documented platforms deleting potential war-crimes evidence (HRW), and Palestinian digital-rights groups report widespread removals of Palestinian content in 2024 (Global Voices). **Availability status should be recorded as a variable**, not silently enforced as deletion. Publication should use a "dehydrated" release of IDs plus codes, with media available to vetted researchers under a data-use agreement.

A symmetric codebook with humans holding the contested variables

Dimensions borrowed from published Israel–Gaza video studies

No validated codebook exists for Israel–Gaza short video. A defensible one can be assembled from published work. The table pairs each dimension with its source and with who should code it. Prevalence figures on stance are themselves contested. A non-peer-reviewed Cybersecurity for Democracy analysis reported about 17 pro-Palestinian videos per pro-Israel video on TikTok, but its labels and their framing in press coverage are disputed (C4D; JNS). That is a reason to define categories symmetrically and pilot them on balanced samples.

Dimension	Categories	Source	Coder of record
Source type	News (state-funded or independent), state or military, politician, NGO or UN, activist, influencer, ordinary user, synthetic	Miehling & Kübler 2026	Human, with automated pre-fill
Stance	Pro-Israeli, pro-Palestinian, both, neither; coded separately for the creator and for the actors portrayed	Common binary-plus-neutral practice; the creator/portrayed split is suggested here	Human (defines conditions)
Framing functions	Problem, cause, moral evaluation, remedy, each attributed to an actor	Entman 1993	Human
War or peace orientation	War, peace, neutral (79.5%, 11.1% and 9.4% of 405 TikTok news videos)	Fahmy et al. 2025	Human, model-assisted
Moral foundations	Care/harm, fairness, loyalty, authority, sanctity	Abbas et al. 2026	Human, model-assisted
Emotion appeal	Anger, fear, sadness or grief, hope, humour	Suggested here; humour functions coded in a study of 257 Israeli TikToks	Model-assisted, human-validated
Graphic content	Casualties, injured people, children, bodies, destruction, plus the platform interstitial flag; an ordinal graphicness rating	Inference from the welfare literature	Human (welfare-critical)
Dehumanisation	Animal or vermin metaphors, collective blame, applied symmetrically to all groups	STAIR lexicon; ADL	Human
Visual scene	Combat, destruction or humanitarian, political, news or interview, protest, symbolic or religious, other	Miehling & Kübler 2026	Vision model, human-validated

Dimension	Categories	Source	Coder of record
Verification status	Known authentic, known synthetic, recycled or out of context, unverified, plus a fact-check URL	Inference from detector-failure evidence	Human

Reliability at α ≥ .800, higher for variables that define conditions

The conventions are to rely on α ≥ .800, to treat .667–.800 as grounds for tentative conclusions only, and to discard variables below .667. The bar should rise as the cost of error rises (Krippendorff; ATLAS.ti manual). In this study the costliest variables are the ones that **assign clips to conditions**: perspective, frame and graphicness. A misclassified clip contaminates the manipulation itself, not just a descriptive statistic. Those variables should clear .800 with a bootstrapped confidence interval, using the R `icr` package for example (CRAN). Reliability samples should be sized by the Bloch–Kraemer formula rather than by rule of thumb; one tabulation gives at least 81 codings for α = .667 with two coders and up to 10 sub-codes (ATLAS.ti). Rare categories such as atrocity imagery need stratified oversampling, or α will be unstable.

Automated coding is a validated assistant, not a coder of record

The evidence on LLM annotation cuts both ways. Gilardi et al. found that zero-shot ChatGPT was about 25 points more accurate than MTurk workers, at under $0.003 per label (PNAS). But Baumann et al. replicated 37 annotation tasks with 18 models and 13 million labels. Annotation choices produced incorrect conclusions for **about one in three hypotheses with state-of-the-art models** and half with small models, and common statistical corrections were "largely ineffective" (arXiv 2509.08825). Validation must be task-specific (Pangakis et al.). The closest domain pipeline studied 2,371 Israel–Gaza YouTube Shorts from Al Jazeera, BBC, DW and TRT World. It used Whisper large-v3 transcription, DeBERTa aspect sentiment (macro-F1 81.9, against 72.5 for a fine-tuned Qwen2.5-7B) and Qwen3-VL-4B scene classification at one frame per second (86.9% accuracy on 799 checked frames). It excluded non-English Shorts and reported no α (Miehling & Kübler).

Language is the weak link. The ivrit.ai Hebrew Whisper models report up to 29% lower error rates than stock Whisper (Interspeech 2025). Dialectal Arabic remains harder, and no Levantine or Palestinian social-video benchmark was found. Tesseract ships Hebrew and Arabic OCR data (Tesseract), but right-to-left and mixed-script overlays, which often carry the stance in Reels, need their own validation. **Deepfake detectors are unreliable on this conflict.** One labelled a genuine Netanyahu video "96.9 per cent AI-generated", and Hany Farid cautions that a "not AI" result does not establish authenticity (Khaleej Times/AFP). Verification should combine platform labels, C2PA credentials where present (C2PA), reverse search for recycled footage and fact-check matches. Detector output counts as one weak signal.

The validation protocol has five steps. Build an adjudicated human gold standard, stratified by language, source type and rare categories. Report per-variable F1 and human–model α, treating the model as an extra coder. Preregister the model, prompt and temperature, and test sensitivity to paraphrased prompts and alternative models. Keep humans as coders of record for stance, dehumanisation, graphicness and verification. Prefer open-weight models run locally, or archive exact API model versions. Running models locally also keeps special-category data away from third-party APIs, a data-sharing risk that the Piccardi et al. methods guide flags (arXiv 2406.19571). Coders and

curators are themselves exposed: AoIR warns that "simply reviewing and curating" extreme violence "can have serious consequences for researchers' psychological health" and requires therapeutic counter-measures (AoIR IRE 3.0).

Fork a video-native research feed instead of building one from zero

Two open tools already log per-clip watch time in a vertical feed

Two open tools already provide a mobile vertical feed with per-clip logging. The older simulators are mature but were designed around text and images.

Tool	Licence	Stack	Short-video fit	Telemetry	Caveats
DICE-TikTok	CC BY-NC-SA 4.0 per its LICENSE file; a secondary summary calls oFeeds MIT (LICENSE)	oTree, oTreeHub deploy, CSV-driven content	Full-screen vertical video, swipe, likes, comments; designed for iOS Safari and Android Chrome (README)	Per participant × video: watch time excluding pauses, liked, comment, sequence position; device data per session	Non-commercial licence; RTL support undocumented
WatchLens	MIT (GitHub)	React/TypeScript, FastAPI, PostgreSQL, RecBole, Docker Compose	TikTok 9:16 and YouTube presets; self-hosted or embedded content; pluggable recommenders; pre-, post- and inter-session surveys	33 event types in 6 categories, each row carrying the recommender and feed position (RecSys '26)	Tested with only 30 students; single server not tested at high concurrency
VidShare (Truman)	"Other" in GitHub metadata; Truman itself is MIT (Truman LICENSE)	Node.js, Docker	Shorts/TikTok-like, with scripted actors, comments and notifications (VidShare)	Standard export has no per-post dwell-time column (Truman docs)	Instrumentation must be added
Mock Social Media Website Tool	Apache-2.0	AWS or local	Video posts, but Facebook and Twitter layouts only (docs)	Reactions by post ID; per-post viewing time listed as a future feature	No vertical swipe
(Mis)Information Game	CC BY 4.0	Node.js, Firebase	"Does not currently support the inclusion of longer videos or sound" (BRM 2024)	Dwell per post in ms	Unsuitable for video
FeedMonitor	No licence listed	Manifest V3 browser extension	Reranks real desktop feeds; blocked on mobile by certificate pinning (GitHub)	Visibility at 1 s and 3 s, heartbeats	Cannot reach TikTok, Reels or Shorts apps

DICE-TikTok is the shortest path for fixed, researcher-curated sequences, provided a non-commercial share-alike licence and oTree suit the team. **WatchLens is the better base if an adaptive-recommender arm is wanted**, since it already separates policies and logs exposure context. Neither documents right-to-left Hebrew or Arabic layouts, which must be tested.

Conditions should hold topic share, runtime and slot position constant

The Hameiri design embedded three 30-second paradoxical clips among three generic commercials in counterbalanced five-minute sessions. Its control was six commercials, three of them shared with the treatment, so topical density was held constant. Fifteen distinct clips were used, and **five of six pilot clips failed**, a reminder that stimuli need piloting (Hameiri et al.). Incerti et al. flagged differing shares

of non-target video across feeds as a confound. They split participants into Forced (no skipping, 20%) and Free Choice (80%) arms to separate dose from choice, drawing on pools of 67 influencer and 89 state-media videos (Incerti et al.). An MIT preregistration randomises which videos appear and in what order at the participant × video level, stratified into thirds by content type (AEA RCT 18673). Demand effects appear limited in online survey experiments (Mummolo & Peterson), so filler masking serves mainly realism.

Arm	Content per session	What it isolates	Evidence and risks
Placebo control	Filler only, runtime matched	Test-retest drift and generic video effects	Animal and entertainment placebos (Landgrave; Incerti)
Topic control (suggested here)	Neutral wire-style conflict news at the same target-to-filler ratio	Topic salience versus persuasive content	Inference: separates "being reminded of the conflict" from the manipulation
Balanced	Equal Israeli- and Palestinian-perspective clips among filler	Cross-cutting exposure	Near-zero effects in the Liu and Nyhan studies; backlash rare in Guess & Coppock but seen in Bail et al.
Humanising	Personal stories from both sides	Dehumanisation and warmth	Contact reduced dehumanisation in a US–Muslim study (Bruneau et al. 2021); "helping hurts" backfire among right-wing Israelis
Paradoxical thinking	Exaggerated agreement with the viewer's own group's conflict-supporting beliefs	Unfreezing	Validated for Jewish Israelis; needs audience-specific stimuli for other groups (inference)
Optional factor: adaptive versus fixed	Same slot constraints; target slots filled by an engagement rule	Algorithmic amplification	WatchLens's similarity policy raised cumulative watch time (medians 1,078.5 s versus 700.2 s) (RecSys '26); attitude effects untested
Optional factor: forced versus free	No skip or a minimum watch time, versus free swipe	Dose versus choice	Incerti's 20/80 split; Liu's 30-second minimum
Optional: prebunking	Short videos teaching curation and sequencing techniques	Resistance to manipulation	About 5% better technique recognition in a YouTube field test at about $0.05 per view (Roozenbeek et al.)

Sequencing rules follow from these precedents. Fix the target-to-filler ratio across treatment arms. Match total runtime and the clip-length distribution. Counterbalance slot positions with a Latin square, so each clip type appears in each position equally often. Draw clips at random within slot types from large pools of equivalent clips, and model clip-level variation. Randomise server-side after the baseline, stratified by identity group and political orientation. Every added factor multiplies the sample needed, so the adaptive, forced and prebunking factors belong in pilots unless the budget supports them. Dose matters too. Incerti et al. estimate about +0.1 favourability per extra minute of influencer content (Incerti et al.), and the only durable effects came from repeated sessions. If durability is a goal, three to six sessions over two to four weeks is the realistic template.

Exposure-aware telemetry with a monotonic clock

Existing tools converge on an event log keyed by participant × clip × position. The log covers playback events, visibility, interactions and the policy that produced each impression (WatchLens; DICE-TikTok). Client timing should use `performance.now()`, which runs on a monotonic clock that "never decreases", unlike `Date.now()`, which follows system clock adjustments (MDN). It should be paired with server UTC timestamps.

Table	Fields
impression	Hashed participant ID, session ID, condition, stratum, policy ID, feed position, clip ID, clip attributes snapshot, sequence seed, server UTC time, client start and end in ms

Table	Fields
playback_event	Play, pause, seek, ended, progress (each quartile or about every 250 ms), media time, client ms; derived: watch time excluding pauses, maximum position, percent watched, loops, time to skip
interaction	Like, unlike, comment, share, mute or unmute, swipe direction
visibility	Tab hidden or visible, app backgrounded, idle, heartbeat
device	User agent, viewport, touch capability, orientation, network type
welfare (added here)	Distress check responses, stop or "this upset me" events, skipped flagged clips, support-link clicks

Self-hosted video, server-side sequencing and equal latency

The recommended build has five parts. The front end is a progressive web app with a full-screen vertical `<video muted playsinline>` carousel. Chrome always allows muted autoplay, and sound needs a user gesture (Chrome). On iOS, `playsinline` prevents forced fullscreen, and unmuting without a gesture pauses playback (WebKit). A "tap to start with sound" screen therefore unlocks audio for the whole session. Video should be pre-transcoded H.264 MP4 or HLS at 540p–720p, served from university object storage or a CDN with signed URLs, with the next two clips prefetched. The back end, FastAPI or Node with PostgreSQL, assigns conditions, generates and logs each seeded sequence and keeps an append-only event table. Participants move from the panel to the baseline survey, the app and the post-survey through redirects that carry a hashed panel ID and a signed session token. The (Mis)Information Game's developers report that return-to-survey linking lost data without cookies, and that one integration error forced eight exclusions (BRM). Latency must be equal across arms: about 2 seconds is tolerated, user attitudes decline beyond about 4 seconds, and about 10 seconds causes disengagement (arXiv 2406.19571).

Recruitment splits into three populations with different channels

Population	Channel	Notes
Jewish Israelis and Arab citizens of Israel	iPanel (over 100,000 members, with a dedicated Arab-sector panel) (iPanel); Midgam; Panel4All; the TAU Cohen Institute probability web panel (about 10,000 Jewish and 1,000 Arab members at the time of its report) (TAU)	Prices are not public. Hameiri et al. paid about 26 ILS for a nine-wave panel and lost about 35% across the exposure month (PNAS)
Palestinians in the West Bank and Gaza	A field partner such as PCPSR, which polls face to face on tablets (N = 1,270 in May 2025) (PCPSR)	The app would run pre-cached on field devices. Gaza interviews took place partly in shelters (PCPSR Poll 96). Exposing war-affected populations to conflict video needs its own risk assessment
International or third-party samples	Prolific (Israel listed, pool size unpublished) (Prolific); Cint/Lucid	Prolific's minimum is £6/$8 per hour and its recommendation £9/$12 (Prolific); a 33.3% academic platform fee is reported by third parties only (CleverX)

Ethics review will be full, and the debrief has to do real work

Every applicable regime points to full committee review

Deliberately shifting attitudes about a live war, with real footage and a concealed persuasive aim, does not fit any exemption pathway.

Regime	Relevant rule	Implication
US Common Rule (if US federal funds or a US IRB are involved)	The "benign behavioral intervention" exemption requires interventions that are "harmless" and that the investigator has "no reason to think" subjects will find "offensive"; it does not apply to children (45 CFR 46.104). Altering consent requires minimal risk and impracticability without the alteration (46.116(f))	Full or expedited review; check whether the Israeli institution holds a Federalwide Assurance
APA Ethics Code	8.07(a) allows deception only where it is justified and no non-deceptive alternative is feasible; 8.07(b) bars deceiving participants about research "reasonably expected to cause... severe emotional distress"; 8.07(c) requires explanation and a chance to withdraw data (APA)	Conceal the direction of persuasion only, never the graphic nature of the content
Israeli university committees	Tel Aviv University's IRB "is not the 'Helsinki Committee'" and covers non-medical research (TAU). All Hebrew University committees have operated to IRB standards since June 2025, with CITI training and Privacy Notice requirements (HUJI)	A Helsinki Committee is needed only if recruiting via medical settings. Schools need Ministry of Education Chief Scientist approval (Herzog College)
EU committees and funders	The Finnish model requires review for any deviation from informed consent, "exceptionally strong stimuli" or mental-harm risk (University of Helsinki). The Horizon Europe ethics self-assessment, including "misuse of results", becomes contractually binding (EUREKA)	EU partners will run their own review
EU AI Act	Art. 5(1)(a) bans "purposefully manipulative or deceptive techniques" that cause significant harm; commentary says the research exclusion does not cover real-world testing on people (Covington)	Get a legal opinion if the adaptive arm uses a learned recommender; a fixed, curated sequence is less likely to count as an AI system at all
AoIR IRE 3.0	Lists "manipulating news feeds" and "intentional triggering" among classic harms; war imagery demands "greater care" (AoIR)	The deliberative standard committees will apply

Consent can conceal the persuasive direction but not the processing or the graphic content

The standard compromise is **authorised deception**. Participants are told up front that "some aspects of the study will not be described until the end", which the Common Rule recognises as prospective agreement (45 CFR 46.104(d)(3)(iii)). The consent form must lead with "a concise and focused presentation of the key information" and say how identifying records are protected and whether de-identified data may be reused (46.116). Three further elements are non-negotiable in this study. First, an explicit statement that political opinions will be collected and analysed, because GDPR Art. 9(2)(a) explicit consent cannot rest on a concealed purpose (GDPR Art. 9). Second, a specific description of any violent or graphic content, given APA 8.07(b). Third, a guarantee that participants can skip or stop any clip without losing pay. The Hebrew University also requires a statement that the study "will collect data about you that can identify you", plus the university Privacy Notice and Data Protection Officer contact (HUJI).

Standard debriefs leave politically charged beliefs in place

Ross, Lepper and Hubbard showed in 1975 that after a standard "outcome" debriefing, "both actors and observers showed substantial perseverance of initial impressions". A "process" debriefing that explicitly explained perseverance largely eliminated the effect (JPSP 1975). Recent work shows standard debriefs fail most for political material. In Clayton et al., an enhanced debrief raised belief

accuracy by more than 2 points on a 7-point scale where the original debrief did nothing, and false beliefs about vaccines and election security persisted after standard debriefs (PNAS Nexus 2024). Hoes and Altay found a claim-level fact-check debrief cut belief in false claims by 0.78 points. A general warning had no effect and slightly reduced belief in true claims (−0.20). Fewer than 30% of 346 misinformation studies they reviewed reported any debriefing (PSRM 2026). Greene and Murphy's thorough debrief cut false memories from 12% to 7.25% at one week, yet 11.52% still held both memory and belief for the stickiest story (PLoS ONE). There is no direct evidence on reversing *attitude* change induced by curated short video. That gap is itself an argument for measuring residual effects after the debrief.

The debrief should therefore have seven parts, symmetric across arms: a dehoax that names the intended direction and the curation and sequencing technique; a process explanation of why impressions persist after their basis is discredited; claim-level corrections, with sources, for any false or exaggerated clip, which covers every paradoxical stimulus; a balanced summary; a wellbeing check with resources in Hebrew, Arabic and English; re-consent and data withdrawal; and a short post-debrief re-measure, repeated with a re-debrief at follow-up.

Welfare protections must assume a traumatised sample

A Tel Aviv University study found 23% of adult Jewish Israelis had PTSD after 7 October (AFTAU). A TAU smartwatch study of more than 5,000 Israelis linked heavy viewing of graphic attack footage to higher PTSD risk, with reported rates of 22.9–36.0% around two months and 15.9–24.7% at seven months (News-Medical). Very high trauma burdens are also reported among Palestinians, but the specific figures found could not be verified against primary papers. A meta-analysis found that **content warnings do not reduce distress** and slightly raise anticipatory anxiety (APS). Warnings serve informed consent; they are not harm reduction, so other protections have to carry the weight. Minors should be excluded or given a separate non-graphic protocol under Ministry of Education rules. Bereaved families, evacuees, released hostages and their families, recently returned reservists, Gaza residents and people with high current PTSD symptoms should be pre-screened and routed out or into non-graphic arms. Graphicness should be capped through the coded rating, with the worst frames blurred, and every clip should carry persistent skip and stop controls. In-feed distress checks need a predefined stop-and-call-back threshold, support resources must be verified in all three languages, and the curation and coding team needs its own support, as AoIR requires.

Political opinions are the most sensitive data the study will hold

Under Amendment 13, political opinions are "highly sensitive" data. A Data Protection Officer is mandatory for large-scale processing of such data, the Privacy Protection Authority can impose penalties in the hundreds of thousands of shekels or more, and courts can award statutory damages up to NIS 10,000 without proof of harm (Herzog Fox & Neeman). A study of a few thousand participants is far below the 100,000-person reporting threshold. A university will still likely require DPO sign-off, a database-definition document and a security procedure under the 2017 Data Security Regulations. EU-to-Israel transfers rest on an adequacy decision the Commission maintained in January 2024 (Hunton). The EDPB criticised the review (EDPB), and civil-society groups have urged reassessment (Statewatch).

The targeting risk is concrete. By 22 October 2023, the legal-rights group Adalah was monitoring 89 disciplinary cases at 25 Israeli institutions over social-media expression and had documented at least 80 related arrests (Adalah). AP later reported more than 270 Palestinian citizens arrested, according to Adalah's advocacy counts (AP via KVIA). Minimisation should therefore go beyond compliance: no

names, webcam or face video, or precise location; age bands and broad regions only; no identifying free text; a panel-ID key held by a single custodian and destroyed after the withdrawal window; hosting on university infrastructure in Israel or the EU with no third-party analytics; and suppression of any published cell with n < 10.

Four controversies set the norms boards will apply

Facebook's 2014 emotional-contagion study manipulated 689,003 News Feeds and claimed the Data Use Policy constituted consent. PNAS issued an Editorial Expression of Concern (PMC). The lesson: platform or panel terms are not consent.

The Meta 2020 election studies were judged rigorous by their independent rapporteur. He also concluded that "independence by permission is not independent at all" (Science). Critics later argued that 63 platform changes overlapped the study window (UMass). The lessons: put independence terms in writing, and log every mid-study change.

Cambridge Analytica harvested data on up to 87 million users through a research-framed app (Computer Weekly). The lesson: political profiles must never be repurposed.

The University of Zurich's 2025 Reddit experiment posted 1,783 AI comments, using personas that included a rape victim and a trauma counsellor, to unconsenting users. Its ethics committee's advice was "not legally binding". The PI was formally warned, and the findings went unpublished (Retraction Watch). Its leaked draft claimed persuasion rates of 13–25% against a 3% human baseline (IGP). Boards will ask how a persuasion study on the war differs from it. The answer has to be real consent, binding conditions, no fabricated identities in stimuli and no unconsenting audiences.

Misuse, funding and directional pre-registration

AoIR asks researchers which actors would use a method "for malevolent purposes" and whether misuse can be contained "by design" (AoIR). The practical answers are to run the app only behind authenticated, consented sessions with no public URL or app-store release, hold stimulus sequences in controlled-access repositories, publish sequencing logic at the level of principle rather than as a turnkey orchestration engine, never share models trained on participants' political data, and run a misuse review before publication.

The Pentagon's Minerva programme shows how defence funding of social science draws objections on independence (Times Higher Education). Funding from any party to the conflict, including government public-diplomacy units, should be declared. A study funded by one side and designed to move opinion toward that side would face severe credibility objections. Preregistering the direction for each arm, the primary outcomes, the debrief-reversal outcomes, stop rules and the stimulus list, ideally as a Registered Report, shows that sequences were not tuned to an advocacy goal.

Lower-risk designs that keep the paradigm

The most defensible variants change the target, not the medium. One option is to aim outcomes at dehumanisation, support for violence against civilians or recognising manipulative editing, rather than at taking a side. Dehumanisation is symmetric and predicts hostility on both sides, which makes it a non-partisan, welfare-relevant target (Bruneau & Kteily). Another is a symmetric design with pro-Israeli-framed, pro-Palestinian-framed, balanced and neutral sequences, which recasts the question as "how does sequencing persuade?". A third is to pair persuasion arms with a prebunking module on curation techniques and use it in everyone's debrief. A fourth is adversarial collaboration, in which

Jewish-Israeli and Palestinian co-leads jointly approve stimuli, hypotheses and the debrief text and preregister competing predictions. The method is established, though time-consuming, and some collaborations fall apart (APA Monitor). Hameiri et al. show that Israeli committees have approved campaigns intended to move conflict attitudes toward conciliation (PNAS). A non-de-escalatory direction would need a much stronger justification.

Conclusion

The evidence reframes what this study can deliver. Its most credible output is a **precise estimate of how much sequencing matters**, including a well-bounded null if that is what happens, not a demonstration that short video changes minds. That estimate has to be measured separately for each population and for left and right, because the two interventions with Israeli precedent, paradoxical thinking and humanising stories, have documented subgroup asymmetries and at least one backfire. Two design decisions carry unusual leverage. Commissioning or licensing matched clips settles the copyright question and the stimulus-equivalence question at once, and lets the team avoid relying on platform research programmes that do not deliver video. Treating per-person KPIs as instruments of quality and welfare rather than of persuasion aligns the measurement plan with what the reliability arithmetic allows.

The ethics case and the scientific case also converge. The safeguards a board will demand produce data the analysis needs anyway: a pre-treatment attention screen, a control arm that calibrates test-retest noise, a post-debrief re-measure and a symmetric design. In a conflict where both sides' humanity ratings of the other sit near zero, a study that reports residual post-debrief effects as a safety outcome contributes something the literature currently lacks. So does a study that demonstrates how to do this research without becoming a persuasion tool for either party.

An ordered implementation plan

Step	Key actions	Gate before the next step
1. Governance	Form a co-led team (Jewish-Israeli and Palestinian researchers, ideally as an adversarial collaboration). Fix the populations and the primary question. Declare funding and sign independence terms. Assign partner roles: a US, EEA, UK, Canadian or Swiss partner as TikTok API principal; an EU research organisation for DSA Art. 40 and DSM Art. 3 work	Written independence terms; decision on whether the design is symmetric
2. Legal and data protection	Obtain a copyright opinion for each stimulus class. Run an EU-side DPIA. Prepare the Amendment 13 database-definition document and security procedure, with DPO sign-off. Get an AI Act opinion if the adaptive arm uses a learned model. Check Federalwide Assurance and IRB reliance if US funds are involved	Written opinions on file
3. Access applications	File for the YouTube Researcher Program and the Meta Content Library; file for TikTok and DSA access through partners; keep the design independent of the outcome	Runs in the background
4. Stimulus sourcing	Commission or license matched clips in Hebrew, Arabic and English. Build the reference corpus with the lowest-risk tools, WACZ archives, the full metadata record, SHA-256 and PDQ/TMK hashes and availability re-checks	Rights basis recorded for every candidate clip
5. Coding and validation	Train coders on the symmetric codebook. Size reliability samples by Bloch–Kraemer, oversampling rare categories. Reach $\alpha \geq .800$ on condition-defining variables. Rate and cap graphicness. Validate the automated pipeline against a gold standard, including prompt-sensitivity tests. Provide counselling for coders	Reliability report; ethics committee approves the clip pool

Step	Key actions	Gate before the next step
6. Instrument	Assemble the battery with Pulse wording and multi-item composites. Translate and back-translate. Add filler items. Run cognitive pretests in each language and test RTL rendering	Pretest findings resolved
7. App build	Fork DICE-TikTok or WatchLens. Add server-side stratified randomisation after baseline; slot plans with a fixed ratio, matched runtime and Latin-square positions; the telemetry schema; welfare controls; panel handoff; equal latency; a security review	Telemetry validated on iOS Safari and Android Chrome; load test passed
8. Ethics submission	Seek full review at the Israeli committee and at partner committees, with consent text, the seven-part debrief, the distress protocol, verified support resources and the misuse assessment	Approvals in writing
9. Pilot	Run per population: clip-level manipulation checks, distress rates, telemetry plausibility, control-arm test-retest for RCI, the baseline–outcome correlation for power, post-debrief residuals	Failing clips dropped; power recalculated
10. Preregistration	Register on OSF or AsPredicted, or as a Registered Report: the direction for each arm, primary composites, the MDE and equivalence bounds, pre-treatment exclusion rules, moderators, stop rules, the stimulus list and seeds	Timestamped registration
11. Fielding	Wave 0 baseline; Wave 1 exposure, post-test and debrief; Wave 2 at one to two weeks with re-debrief; optional Wave 3. Monitor quality and welfare dashboards and freeze the change log	Stop rules applied throughout
12. Close-out	End the withdrawal window, destroy the linking key and archive data under controlled access	Destruction logged
13. Analysis and release	ANCOVA by population, preregistered heterogeneity tests, equivalence tests, descriptive RCI, residual post-debrief effects. Release the corpus dehydrated, after a misuse review	Publication

What should change in the Exposure Feed Lab prototype

The prototype at `prototypes/feed-lab/exposure-feed-lab.html` already does several things the evidence recommends. It draws clips reproducibly from seeded, typed slots (`planSequence`), assigns conditions in permuted blocks, logs each impression with `performance.now()` timing, uses authorised-deception wording at consent, offers a no-data decline path and a withdrawal button, and warns in the console that per-person change is noisy. Its gaps fall into two groups. Some are measurement bugs that would corrupt data. Others are design gaps that the research above identifies. Priority 1 items should be fixed before any pilot, and Priority 2 items before fielding.

Area	As built	Change	Priority
Slider defaults	`renderSurvey` writes each slider's default into the answers on render (thermometers 50, humanity 100), so an untouched slider is recorded as a real answer and the "missing" check never fires	Show no handle until the participant interacts, store a "touched" flag and require a response. A default of 100 on humanity anchors at the ceiling in a population where 42–71% answer zero	1
Watch-time telemetry	The replay handler jumps `watched` to the end of the current loop, so pressing ↺ after 3 seconds logs 100% completion plus a replay. rAF ticks pause in hidden tabs, and the first frame on return adds the whole gap to `watched` and `dwellMs`	Drive watch time from real `<video>` events; count replays only on genuine loops or seeks to zero; add `visibilitychange` handling, foreground-only watch time and a heartbeat	1

Area	As built	Change	Priority
Console aggregates	The by-condition table includes EXAMPLE sessions and treats missing values as zero (`x[f] ?? 0`); KPI colours mark every increase green and every decrease red, so a falling warmth gap (less polarisation) shows red and rising annexation support shows green	Exclude examples, handle missing values properly, use neutral non-valenced colouring, report results by identity group, and show ANCOVA estimates with confidence intervals instead of raw mean changes	1
Randomisation timing	`assignCondition` runs in `newSession` before consent, so declines consume block slots; the blocks are unstratified; researchers can force a condition	Randomise server-side after baseline, stratified by identity group × political orientation; restrict forcing to a logged test mode	1
Identity and moderators	Three options, with Palestinian citizens of Israel grouped with Palestinians elsewhere; no political orientation	Separate Jewish Israelis, Arab citizens of Israel, Palestinians in the West Bank/East Jerusalem, Gaza and the diaspora, and third parties (the Pulse shows Arab citizens differ sharply); add left–right orientation, religiosity and age band	1
Outcome items	Generic "Israelis" and "Palestinians" thermometers; an "Ascent-of-Man style" item with a "Less" anchor; unqualified "annexation"; single-item hope	Target-specific thermometers; "How human are [target]?" with validated anchors, scored ingroup minus outgroup; Pulse solution wording including "without equal rights" and the Palestinian mirror option; 3–4-item hope; add trust, parochial empathy, fear, unfreezing and knowledge items, with fillers	2
Paradoxical arm	The same exaggerated clips go to every identity group; the expected manipulation-check answer is n/a	Use audience-specific paradoxical stimuli, or restrict the arm to Jewish Israelis for whom it is validated; add a perceived-extremity check; pilot every clip, since 5 of 6 of Hameiri's pilot clips failed	2
Condition balance	Target:filler slots are 0:8, 5:3, 5:3 and 4:4; full-view runtime is about 170 s (control), 242 s (balanced), 224 s (humanising) and 167 s (paradoxical), computed from the placeholder codebook; slot positions are fixed within each arm; the control description says it "gives the baseline for… pretest sensitisation"	Fix the ratio across treatment arms, match runtime, counterbalance positions with a Latin square, enlarge clip pools, and consider a topic-control arm. Reword the control description: with a pretest it estimates test-retest drift only, and measuring sensitisation would need a post-only arm, which Clifford et al. suggest is unnecessary	2
Adaptive mode	A researcher toggle rather than a randomised factor; the engagement score pools filler clips, so in the balanced arm a "calm" filler steers target picks toward "calm" history explainers; nothing limits escalation toward anger, fear or grief	Randomise adaptive versus fixed; score target clips only; cap emotional intensity and graphicness; log a policy ID per impression; obtain an AI Act opinion if the rule becomes a learned model	2
Waves	Pre and post in one sitting, about four minutes apart; no follow-up	Baseline wave a week earlier, immediate post-test, one-to-two-week follow-up and a post-debrief re-measure	2
Data-quality flags	`attnOk` requires both the pre- and post-treatment checks; a post-survey under 20 seconds is flagged	Exclude on pre-treatment checks only; report post-treatment checks; never use timing as Prolific rejection grounds; add telemetry plausibility flags	2
Welfare controls	"Next" is the only control in the feed; withdrawal is offered only at consent and debrief	Persistent "Stop" and "This upset me" buttons, in-feed distress checks with a threshold protocol, and blur or opt-in for flagged clips	1

Area	As built	Change	Priority
Consent screen	Generic "may be upsetting" warning and placeholders for contact and approval details	Key information first; explicit consent to processing political opinions; a specific description of graphic content; Privacy Notice and DPO; funders; withdrawal window and code; vulnerability screen; choice of Hebrew, Arabic or English	1
Debrief	States the condition and that "no platform algorithm was involved", even in adaptive sessions; placeholder helplines; local deletion only	Correct the statement for adaptive sessions; implement the seven-part debrief with claim-level corrections for paradoxical clips; verified tri-lingual resources; server-side withdrawal by code; re-debrief at follow-up	1
Per-person KPI cards	Raw changes on eight attitude items	Show RCI flags calibrated on the control arm, plus exposure, quality and welfare KPIs; restrict console access to need-to-know	2
Clip codebook	11 fields; `graphic` is always false; no provenance or rights data	Add the provenance, integrity, rights, stance (creator versus portrayed actors), verification and graphicness-rating fields above, with coder IDs, α and subtitle tracks	2
Storage and languages	localStorage, an English-only interface, Hebrew and Arabic clips without subtitles, Google Fonts requests	A server-side append-only store with salted-hash IDs and a separate key; a full tri-lingual right-to-left interface with matched subtitles; self-hosted assets with no third-party requests	2